DEVELOPMENT OF A GIS-BASED HEURISTIC DECISION SUPPORT SYSTEM WITH A PARTICIPATORY APPROACH IN URBAN REGENERATION



1 PROJECT OVERVIEW

A web-based Spatial Decision Support System (SDSS) has been developed to integrate spatial data management, real estate valuation, negotiation, and planning processes in urban regeneration. Combining Geographic Information Systems (GIS), multi-criteria decision-making, and heuristic optimization methods, the system facilitates interaction among local authorities, experts, citizens, and contractors within a shared digital platform. Piloted in Esenler, Istanbul, the project aims to contribute to data-driven, transparent, participatory, and climate-resilient urban regeneration processes.

2 WHY IS IT NEEDED?

Fragmented data
Spatial, economic and social information remains dispersed.

Complex decisions
Multiple criteria and alternatives require joint analysis.

Limited participation
Stakeholders need transparent ways to take part.

Climate-related pressures
Environmental indicators should inform regeneration.

3 ONE PLATFORM, FOUR MODULES

01 DATA
Integration of spatial, textual and environmental GIS data.

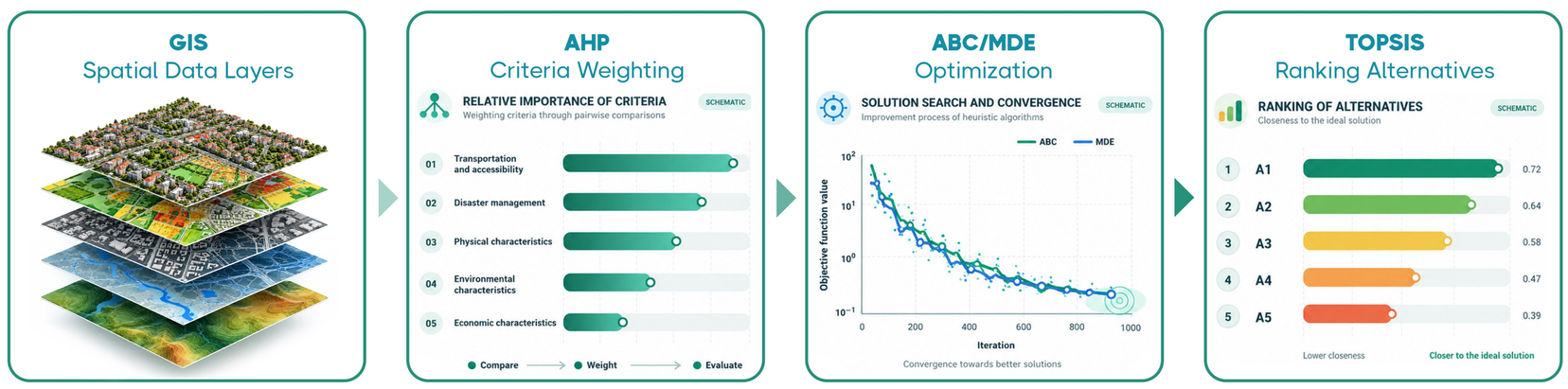
02 VALUATION
Multi-criteria analysis and property value assessment.

03 PARTICIPATION
Stakeholder engagement, evaluation and consensus.

04 PLANNING
Spatial evaluation of urban regeneration scenarios.

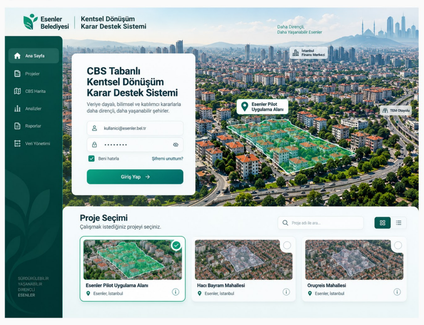
4 SCIENTIFIC METHODOLOGY

The system combines GIS-based spatial analysis, the Analytic Hierarchy Process (AHP), heuristic optimization using the Artificial Bee Colony (ABC) and Multi-population Differential Evolution (MDE) algorithms, and TOPSIS-based multi-criteria decision analysis. Expert-derived criteria weights are optimized to improve consistency and support comparisons among alternatives and planning scenarios.

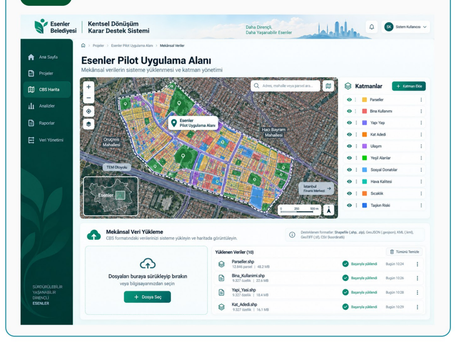


5 PILOT APPLICATION: ESENLER, ISTANBUL

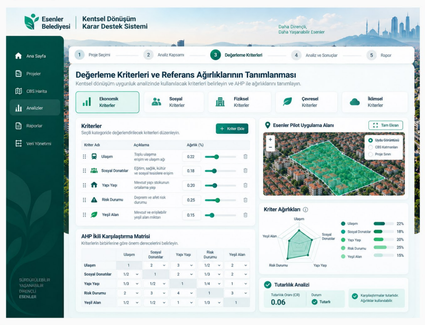
01 System access and project selection.



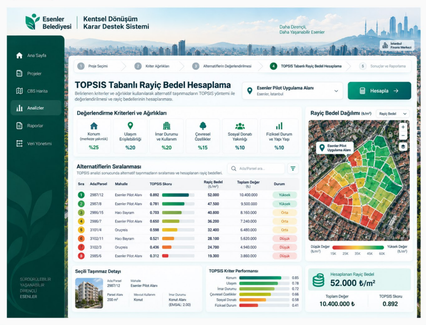
02 Uploading spatial data layers



03 Defining valuation criteria and reference weights



04 TOPSIS-based property valuation and calculations



05 Mapping pre- and post-regeneration data



6 CONTRIBUTION TO CLIMATE-RESILIENT CITIES

Aligned with COP31's Climate-Resilient Cities theme, the system brings air quality, temperature, rainfall, solar exposure, green space, ecological corridors, energy and waste management, and flood risk into GIS-supported regeneration decisions. Its participatory design supports climate adaptation and stronger local decision-making capacity.

Environmental Sustainability

Participatory Governance

Urban Resilience

7 OUTPUTS AND POTENTIAL IMPACT

Technological output
a working web-based Spatial Decision Support System integrating four modules.

Pilot application
Esenler Belediyesi ile saha testleri ve kullanıcı değerlendirmeleri

Scientific output
Articles, international conference papers, book chapters, and postgraduate theses

Scaling potential
an adaptable and expandable decision-support infrastructure for other municipalities and regeneration areas.

2	1	2	1	2
International-indexed journal article	TR Dizin journal article	International conference papers	Book chapter	Master's theses

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